KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2023
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define machine learning and discuss its different types. (06) [CO1]
   b) Derive the Lagrangian dual objective function and formulate the equations that uniquely characterize the solution to both the primal and dual problems for the Support Vector Machine (SVM). (14) [CO3]
   c) Explain how Directed Acyclic Graph (DAG) SVM addresses the drawback of the 1-vs-1 SVM approach, where the number of SVMs increases super-linearly with number of classes, in data classification. (08) [CO2]
   d) Discuss the relationship between Gini Impurity and Information Gain. (07) [CO2]

2. a) How does a separating hyperplane that perfectly classifies all of the training observations lead to sensitivity to individual observations? Explain in detail. (10) [CO2]
   b) Can SVM be over trained? What can be the effect of over training? How can over-fitting be detected for a SVM model? (07) [CO2]
   c) Consider the following two hyperplanes: (08) [CO3]

$$5 + 2x_1 + 5x_2 \quad \text{(for classifier 1)}$$
$$5 + 20x_1 + 50x_2 \quad \text{(for classifier 2)}$$

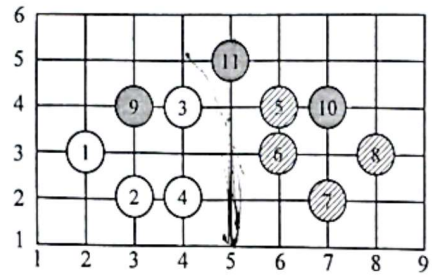
Compute the distance error function and identify which classifier built with these hyperplanes will have better performance.
   d) 'The maximal margin hyperplane for SVM depends directly on only a small subset of the observation' – justify the statement. (10) [CO1]

3. a) Explain Occam's razor principle. In the context of decision tree learning, when we prioritize simplicity over complexity, do you believe Occam's razor adequately justifies the algorithm's inclination toward simpler hypotheses? (10) [CO1]
   b) Discuss the impact of outliers on the performance of a maximum margin classifier. How can the classifier be sensitive to outliers and what techniques can be employed to address this issue? (12) [CO2]
   c) Consider the following table where attribute 'Status' represents class label. Find the root attribute of the decision tree using Gain Ratio as the attribute selection measure. (13) [CO3]

Department	Age	Status	Salary
Sales	31–35	Senior	Medium
Sales	26–30	Junior	Low
Marketing	36–50	Senior	High
Secretary	26–30	Senior	Medium
Secretary	36–50	Senior	High
Sales	36–50	Junior	High
Marketing	31–35	Senior	Medium
Marketing	26–30	Junior	Low
Secretary	31–35	Junior	Medium
Secretary	26–30	Junior	Low

4. a) Discuss the empty space phenomenon and demonstrate the working principle of kernel Principle Component Analysis (PCA) in dimensionality reduction for linearly inseparable input data. Under what conditions does kernel PCA function similarly to standard PCA? (11) [CO1]
   b) What is inductive bias in decision tree learning and which type of inductive bias is used in BFS-ID3 algorithm? Illustrate how BFS-ID3 algorithm exhibits the bias 'shorter trees are preferred over longer trees' with detailed explanation. (12) [CO2]

c) Consider the following training data: (12) [CO3]



Let $y_A = -1, y_B = +1$ be the class indicators for both classes where

$A = \{(1), (2), (3), (4)\}$ and $B = \{(5), (6), (7), (8)\}$

i) Just using the above standing plot, specify which of the points should be identified as support vectors.

ii) Draw the maximum margin line which separates the classes (you don't have to do any computations here). Write down the normalized normal vector $w \in \mathbb{R}^2$ of the separating line and the offset parameter $b \in \mathbb{R}$.

iii) Consider the decision rule: $H(x) =< w, x > +b$. Now determine the class for the points (9), (10) and (11).

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Suppose, we want to fit a line $y = mx + b$ through the points $(1, 2), (2, 3), (3, 4), (4, 5)$ and $(5, 6)$. Assume, initial values of m and b are 0.1 and 0.2 respectively. Calculate the initial error using mean squared error loss function and then apply a gradient descent step. Use learning rate, $\alpha = 0.01$ and calculate the error of fit again after the gradient descent step. (12) [CO3]

b) Demonstrate automatic differentiation (AD) using a suitable example. How does AD differ from numerical differentiation? (07) [CO1]

c) Explain why Convolutional Neural Network is preferred over Multi-Layer Perceptron for image related tasks. (04) [CO2]

d) Explain how artificial neural networks can be used to solve following problems. (12) [CO4]

i) Duplicate/ near duplicate image identification.

ii) Anomaly detection.

iii) Dimensionality reduction.

6. a) Briefly explain the idea of transfer learning. (05) [CO1]

b) What is exploration and exploitation in swarm intelligence algorithms? Explain how these features are provided in following algorithms. (16) [CO1]

i) Genetic algorithm (GA).

ii) Particle swarm optimization (PSO).

iii) Ant colony optimization (ACO).

iv) Artificial bee colony (ABC).

c) What are the disadvantages of Roulette Wheel selection? (03) [CO3]

d) Create a child using Cycle crossover and another child using Enhanced Edge Recombination for the following parents. (11) [CO3]

Parent 1	1	2	3	4	5	6

Parent 2	5	6	1	3	4	2

7. a) "PSO cannot be used for discrete optimization problem" – justify this statement. (05) [CO1]

b) Perform one iteration of PSO to minimize the function $f(x, y) = (x - 2)^2 + (y - 3)^2$ in the range $-10 \leq x, y \leq 10$. Parameters and initial values are given below where the symbols have their usual meanings. (10) [CO3]

Parameters:

$N = 2, \omega = 0.5,$
$C_1 = 1.5, C_2 = 1.5$
$r_1 = 0.3, r_2 = 0.7$

Initial values:

Particle 1: Position $(x_1, y_1) = (5, 5)$,
Velocity $(v_{1x}, v_{1y}) = (0.1, 0.1)$.
Particle 2: Position $(x_2, y_2) = (3, 1)$,
Velocity $(v_{2x}, v_{2y}) = (0.2, 0.2)$.

c) Write down the equations of transition probability and pheromone update of ACO. Also, describe the effects of key parameters. (10) [CO1]

d) Explain the employee bee phase, on looker bee phase and scout bee phase in ABC optimization algorithm. (10) [CO1]

CamScanner

8. a) Explain how clustering can be used for image color quantization. (04) [CO3]
   b) Explain how to calculate and interpret Silhouette score. (06) [CO2]
   c) What is the problem of random centroid initialization in case of k-means clustering? Explain how k-means++ or GA can be used to solve this problem (explain both approaches). (11) [CO2]
   d) Using agglomerative hierarchical clustering create a dendrogram for the data shown in following table. (14) [CO3]

Student_ID	Marks
1	15
2	12
3	33
4	25
5	40

CamScanner